Digital Smile Design (DSD)

The dental domain explained for software developers

Everything you need to know about teeth, smiles, faces and clinical workflow to build an AI-assisted smile design application without making clinical mistakes.

How to read this guide. Part 1 gives the big picture. Parts 2-6 define every dental concept the software touches, with numbers you can code against. Part 7 translates the domain into landmarks, data models and rules. Part 8 lists safety limits. Part 9 is an A-Z glossary. Numeric values are widely used clinical norms and **guides, not absolute rules**: expose them as configurable parameters, never hard-code them as truth.

1. The big picture: what DSD actually is

DSD is a planning and communication workflow, not a single algorithm. A dentist photographs a patient's face and teeth, then draws guide lines and proposed tooth shapes over those photos. The goal is to design a smile that fits the person's **face, lips, gums and teeth together**, show it to the patient before any treatment, and hand the plan to the lab and clinician.

Think of it as **'Photoshop with clinical rules'**: the picture is the input, the dentist's measurements and proportions are the rules, and the output is a proposed new smile plus a set of numbers the lab can build from.

Dental word	Developer translation
Smile design	Fitting a set of 'virtual teeth' to a face under geometric and aesthetic constraints
Facial references	Fixed lines/points on the face (eyes, nose, chin) used as the coordinate frame
Dental references	Lines/points on the teeth and gums (midline, edge line, gum line) that must agree with the facial frame
Mock-up	A temporary physical preview of the proposed smile placed in the patient's mouth
Wax-up	The 3D sculpted model (physical or digital) of the proposed teeth
Restoration	Anything that replaces or covers tooth structure: veneer, crown, filling, bridge
Case	One patient treatment plan = photos + scans + design + notes

The end-to-end clinical workflow (your app covers the shaded steps)

#	Step	What happens	Data involved
1	Consultation	Patient says what they dislike (color, shape, gaps, gums, crooked teeth)	Free text, goals
2	**Photo and video capture**	Standard photo set under a strict protocol (see 4.1)	JPG/PNG, short video
3	**2D analysis**	Draw reference lines, calibrate scale, evaluate proportions	Landmarks, lines, ratios
4	**Digital design**	Propose new tooth shapes over the photo	Tooth outlines/templates
5	**Patient preview**	Show before/after to the patient	Overlay image
6	3D scan	Intraoral scanner or impression gives 3D teeth	STL/PLY/OBJ
7	3D digital wax-up	Lab or software converts 2D design to 3D	STL, matching 2D to 3D
8	Mock-up / trial smile	Printed or molded preview tried in the mouth	Physical
9	Preparation and final	Teeth minimally reduced (if needed); veneers/crowns made and bonded	Prep guides, CAD/CAM files

2. Tooth anatomy and numbering

2.1 Which teeth matter in DSD

Adults have 32 teeth (28 without wisdom teeth). Smile design focuses on the **maxillary (upper) anterior teeth**: the 6 front teeth from canine to canine, visible when smiling. Often the premolars (4th and 5th teeth from the center) are included for wide smiles.

Tooth	Nickname	Role in the smile	Typical crown size (mm)
Central incisor	'Number 1'	Dominant tooth. Defines the smile's character	Width 8.5-9.0, Length 10-11
Lateral incisor	'Number 2'	Smaller, more variable. Often the tooth with defects or missing	Width 6.0-7.0, Length 8.5-9.5
Canine	'Number 3', cuspid	Corner of the smile. Pointed. Supports the lips	Width 7.0-8.0, Length 10-11
First premolar	'Number 4'	Seen in wide smiles. Blends into the back	Width 6.5-7.0
Second premolar	'Number 5'	Last tooth usually visible in a broad smile	Width 6.5-7.0

Left/right convention: in clinical terms the patient's right is on the viewer's left in a frontal photo. Always label sides from the **patient's** perspective in data and UI.

2.2 Numbering systems (your app must support at least FDI)

System	How it works	Example: upper right central incisor
FDI (ISO 3950), used in Egypt/Europe/most of the world	2 digits: quadrant (1-4), then tooth (1-8) from the midline outward. Quadrants: 1 upper right, 2 upper left, 3 lower left, 4 lower right	11 (the upper left central is 21)
Universal (USA)	Numbers 1-32 starting at upper right third molar, going around to lower right	8 (upper left central is 9)
Palmer notation	Symbol with quadrant bracket plus number 1-8	Numeral 1 inside the upper-right bracket

Upper anterior teeth in FDI: 13 12 11 | 21 22 23 (right canine, right lateral, right central, left central, left lateral, left canine).

2.3 Parts of a tooth surface (names used in every design tool)

Term	Meaning
Crown	The visible part of the tooth above the gum
Root	Part inside the bone. Not visible and not designed in DSD
Incisal edge	Biting edge of front teeth. THE key line in smile design
Labial / buccal surface	Outer surface facing the lips/cheeks (the one seen in photos)
Lingual / palatal surface	Inner surface facing the tongue/palate
Mesial / distal	Side of tooth toward the midline / away from the midline
Cervical	Near the gum line (neck of the tooth)
Gingiva	The gum. Gingival margin = the gum's edge hugging the tooth
Interdental papilla	Small triangle of gum between two teeth
Embrasure	Small V-shaped gap between two neighboring teeth (incisal, at the edge; gingival, at the gums)
Contact point / area	Where neighboring teeth touch each other
Enamel / dentin	Hard outer layer (translucent, glossy) / yellowish layer beneath. Both determine color

3. The face and lips: reference system

Before drawing a single tooth, the dentist builds a **coordinate system** on the face. Every tooth decision is judged against it. For your AI, this is the landmark detection layer.

3.1 Facial reference lines

Reference	Definition	Why it matters	Software note
Interpupillary line	Line joining the centers of both pupils	Horizontal reference. The edge line of the teeth should be roughly parallel to it	Detect both pupils, compute the angle
Facial midline	Vertical line through glabella (between brows), nasal tip and philtrum center and chin	Reference for the dental midline	Perpendicular to the interpupillary line (do not just trust image vertical)
Dental midline	Line between the two upper central incisors	Should ideally match the facial midline. Small shifts (under ~2 mm) are often tolerated; TILT matters more than shift	Report shift in mm and tilt in degrees
Commissural line	Line joining the two mouth corners	Should be parallel to interpupillary line	Detect lip corners
Camper's plane / occlusal plane	Plane from nose base (ala) to ear (tragus). Occlusal plane = plane where upper and lower teeth meet	Orientation of the whole smile. In a frontal view, tilt means 'canted'	Hard to see in a frontal photo; use profile photo
Horizontal reference	True horizontal (gravity), not the image edge	Heads tilt in photos. Rotate the image so the interpupillary line is level before judging	Use head pose / pupil line to normalize

3.2 Lip and smile terminology

Term	Meaning	Normal / target
Lip at rest	Lips relaxed, teeth slightly apart	Upper incisor shows about 1-3 mm in younger adults (less with age). Reveals incisal edge position
Social (posed) smile	Natural-looking smile	Upper lip at about the gum line; teeth visible from canine to canine, often premolars
Full / maximum smile	Big smile	Shows the maximum gum and teeth. Used to evaluate gummy smile
Smile arc	Curve of the upper incisal edges relative to the curve of the lower lip	Ideal = consumer ('consonant'): the upper edges follow the lower lip curve. Flat/reverse arc looks old or less attractive
Gummy smile	Excess gum shown when smiling	Over about 3 mm gum show is considered excessive by many clinicians; 0-2 mm is generally accepted
Buccal corridor	Dark space between the teeth corners and the cheek in a wide smile	Small, balanced corridors look fuller. Too large looks narrow; no corridor looks 'denture-like'
Lip line (high/medium/low)	How much tooth/gum the lip shows	High = full gum shown. Medium = tip of papilla. Low = only part of the teeth
Philtrum / upper lip length	Distance nose base to upper lip edge	Determines how much tooth is naturally visible

Developer takeaway: The same teeth can look excellent or terrible depending on lip position. A design done on a **resting-lip photo** and a design done on a **smiling photo** answer different questions. Store the photo type with every landmark set.

4. Photographs, capture protocol and calibration

4.1 The standard photo set

View	What it shows	Used for
Full face, rest	Whole face, lips relaxed	Facial midline, lip length, rest display
Full face, social smile	Natural smile	Smile arc, buccal corridors, midline
Full face, big smile	Maximum smile	Gum display
Retracted frontal	Teeth only, lips held away with cheek retractors	Detailed tooth proportions and gum margins, the main design canvas
Retracted lateral (left/right)	Side view of the teeth	Tooth inclination, canine relationships
Occlusal (mirror) views	Upper/lower arches from above/below	Arch shape, tooth rotations
Profile (side face)	Side of the face	Lip protrusion, nose-lip-chin balance (E-line), tooth angle
Phonetic images ('M', 'F/V', 'S')	Patient saying the sounds	See 4.3: functional validation of tooth length and position
Shade photo	Teeth with a shade tab next to them	Color communication to the lab

4.2 Calibration: converting pixels to millimeters

Photos have no real-world scale. DSD calibrates by **measuring something known**: for example the dentist marks the real length of a central incisor (taken from the patient's model, 10.5 mm for example) or the width of the reference ruler/object in the photo. The software then derives a **mm-per-pixel factor** and expresses all later measurements in millimeters. Lens distortion and camera angle make calibration approximate: show uncertainty, and never present measurements as exact.

4.3 Phonetics (speech-based checks)

Sound	What should happen	What it validates
'M' or relaxed 'Mississippi'	Lips at rest. Incisal edge slightly visible	Resting incisor display (lip support)
'F' / 'V'	Upper incisal edges lightly touch the wet-dry border of the lower lip	Correct incisal edge length and position (a classic test)
'S' / 'Sixty-six'	Upper and lower front teeth come close, with about 1 mm gap	Correct vertical dimension and edge position

5. The aesthetic rules your AI will encode

DSD borrows from art and from clinical research. These are **guidelines clinicians discuss and sometimes disagree about**; ship them as presets the dentist can override.

Concept	Definition	Typical target values
Width-to-length ratio (central incisor)	Crown width divided by crown height	About 75-80% (e.g. 8.5 mm wide / 11 mm long = ~77%). Under ~70% looks long, over ~85% looks square/short
Golden proportion	Each tooth seen from the front is 0.618 times the width of the tooth next to it going toward the back	Central 1.0 : lateral 0.618 : canine 0.382
RED proportion	'Recurring Esthetic Dental' proportion: each tooth is the SAME percentage of the one before it	Often 70-80%. More flexible than golden proportion; many dentists prefer it
Golden percentage	Apparent widths as % of the total width of six anterior teeth	Central about 25%, lateral about 15%, canine about 10% each side (as seen from the front)
Central vs lateral vs canine incisal edge height	Edges form a stepped line	Central and canine edges roughly at the same level; lateral about 0.5-1 mm shorter (classic 'Gull wing' pattern)
Gingival zenith	The highest point of the gum curve on each tooth	Central and canine zeniths roughly level; lateral zenith about 1 mm lower. Zenith slightly distal to the tooth's long axis
Axial inclination	Tilt of the tooth's long axis toward the midline at the top	Increases progressively from central to canine (teeth lean mesially at the crowns' tips toward the midline)
Incisal embrasures	Gaps at the biting edge between teeth	Get larger moving away from the midline
Connector zone	The length of the contact area between two neighboring teeth	Central-central longest (about 50% of the central crown height); progressively shorter toward the back
Tooth shape (outline)	Square, ovoid (rounded), triangular	Usually chosen to match the face shape or patient's personality. Women often skew rounder, men squarer: avoid stereotyping in AI
Symmetry	Left/right matching	Perfect symmetry looks artificial; allow slight natural variation
Smile line vs lower lip	See smile arc	Edges follow the lower lip curve and touch or nearly touch it
Gingival margin symmetry	Gum lines of matching teeth line up	Right and left centrals symmetrical; difference over about 1 mm noticeable

Warning for AI: Never design from proportions alone. A tooth set that is mathematically perfect can still look wrong if the face is asymmetric, the lips are short, or the patient wants a more natural look. Always provide **the measurement + the reason + a way to override**.

5.1 Color (shade) vocabulary

Term	Meaning	Note
Hue	The basic color family (yellowish, reddish, greyish)	In VITA Classic: letters A (reddish-brown), B (reddish-yellow), C (grey), D (reddish-grey)
Chroma	Color saturation / intensity	In VITA Classic: number after the letter, 1 (low) to 4 (high)
Value	Brightness (light to dark)	The most important component of how teeth look
Translucency / opacity	How much light passes through, notably at the incisal edge	Young teeth are more translucent at the edge
VITA Classical	16-shade guide: A1-A4, B1-B4, C1-C4, D2-D4	Most common

Term	Meaning	Note
VITA 3D-Master	Value-chroma-hue guide with codes like 2M2	Systematic
Bleach shades (BL1-BL4, 0M1-0M3)	Extra-light shades for whitened teeth	Common in current requests ('Hollywood white')
Whitening / bleaching	Lightening natural teeth chemically	Done BEFORE final shade selection because veneers cannot be whitened afterward

6. Treatments and occlusion: what the design leads to

6.1 Treatment options the plan may trigger

Treatment	What it is	When used / what the software should consider
Whitening (bleaching)	Chemical lightening	Color only. No shape change
Composite bonding / direct mock-up	Tooth-colored resin sculpted directly on teeth	Small shape changes. Reversible, cheaper, used for a 'trial smile'
Veneer (porcelain/ceramic, lithium disilicate e.max)	Thin shell (0.3-0.7 mm) bonded on the front of the tooth	Most typical DSD outcome. Needs little or no tooth reduction
Crown (cap)	Full covering of a tooth	Heavily damaged teeth or root-canal-treated teeth
Bridge	Replacing missing teeth by joining crowns	Missing tooth case
Implant	Titanium or zirconia root replaces a missing tooth	Missing teeth. The design guides implant position (prosthetically driven planning)
Orthodontics (braces, clear aligners)	Moving teeth	Used if crowding/spacing is too big for veneers. Often better, because it keeps natural tooth
Gingivectomy / crown lengthening	Surgical gum reshaping/bone adjustment	For gummy smile or uneven gum lines
Lip repositioning	Surgery that limits how far the lip lifts	Severe gummy smile
Enameloplasty / contouring	Slight reshaping of enamel	Minor shape corrections
Full-mouth rehabilitation	Rebuilding many teeth with a new bite	Worn teeth. Needs the bite (occlusion) planned

6.2 Occlusion (how the teeth meet)

Beautiful teeth that bite badly will break. Even a front-teeth DSD must respect these concepts, as the AI might need to flag them:

Term	Meaning	Why it matters in DSD
Occlusion	The way the upper and lower teeth contact	Foundation of long-term success
Overjet	Horizontal distance between upper and lower incisors (normal about 2-3 mm)	Lengthening or pushing out front teeth changes it
Overbite	Vertical overlap of the upper over the lower incisors (normal about 2-3 mm, about 20-30%)	Same
Angle Class I / II / III	Relationship of the first molars/jaws. Class I normal, Class II upper teeth forward, Class III lower teeth forward	Affects what is possible without orthodontics or surgery
Anterior guidance	The front teeth guide the jaw during sliding movements and protect back teeth	Design must not create interferences
Canine guidance	Canines guide the side-to-side movement	Canine tip shape is functional
Vertical dimension of occlusion (VDO)	Jaw closing height	Changing tooth length is limited if VDO is kept constant
Centric relation / maximum intercuspation	Reference jaw position / best fit of teeth	Used for scans and mounting
Parafunction / bruxism	Grinding or clenching	High risk for veneers and ceramics. Might need a night guard
Diastema	Gap between teeth, usually between the central incisors	Closing it changes tooth width ratios
Crowding / spacing / rotation	Tooth position problems	Pure veneers may over-bulk teeth: orthodontics may be preferred

7. Translating the domain into software

7.1 Landmarks the AI should detect

Group	Landmarks	Purpose
Face	Pupil centers (L/R), glabella, nasal tip/base, philtrum, mouth corners (commissures), chin point, face outline	Build the facial coordinate frame, rotation normalization, midline
Lips	Inner lip contour (upper and lower), lip corners	Define the mouth opening where teeth are visible ('smile window')
Teeth	Per tooth: incisal edge points, mesial/distal contact points, tooth long axis, outline/mask, tooth number (FDI)	Measure widths, lengths, ratios, axis
Gums	Gingival zenith per tooth, papilla tips, gingival margin curve	Gum symmetry, gummy smile, crown lengthening need
Midlines	Facial midline, dental midline (upper and lower)	Shift and cant analysis
Smile arc	Curve through the incisal edges and curve of lower lip	Consonant/flat/reverse classification

7.2 Derived measurements (compute these, with units)

- **Midline deviation:** dental midline vs facial midline, in mm and tilt in degrees.
- **Edge-line cant:** angle between the incisal edge line and the interpupillary line.
- **Width/length ratio** per anterior tooth, and the **proportion between neighbors** (golden, RED, golden percentage).
- **Incisal display at rest** (mm) and **gingival display at full smile** (mm).
- **Smile arc type** (consonant, flat, reverse) as a classification with confidence.
- **Buccal corridor ratio:** corridor width divided by the total smile width.
- **Gingival zenith heights and symmetry** (mm differences left to right).
- **Tooth axis inclination** per tooth (degrees).

7.3 Suggested data model (simplified)

Entity	Key fields
Patient	id, age range, sex (optional), consent flags (photo use, AI processing, data retention)
Case	id, patient id, goals (text), status, created by, clinician id
Photo	id, case id, **view type** (rest/smile/retracted/profile/phonetic...), file, calibration (mm-per-pixel + method), capture notes
LandmarkSet	photo id, landmarks (name, x, y, confidence), source (AI or manual), **version**
Measurement	photo id, name, value, unit, normal range used, status (within / outside)
ToothDesign	case id, FDI number, outline (polygon), width mm, length mm, shade, template id, source (AI/manual), version
Scan3D	case id, file (STL/PLY/OBJ), arch (upper/lower), units, orientation
Proposal	case id, design versions, before/after images, clinician approval status (required before showing patients as 'final')
AuditLog	who changed what and when, AI model version used

7.4 File formats

Format	What it is	Use
JPG/PNG/HEIC	2D photos	Analysis, overlays. Keep originals untouched; store EXIF orientation
STL	3D mesh, geometry only (no color)	Printing, CAD/CAM, lab exchange
PLY / OBJ	3D mesh with color/texture	Intraoral scanner output with color
DICOM	Medical imaging format for CBCT (3D X-ray)	Bone/root planning, implants. Handle with strict privacy
CBCT	Cone-beam CT scan	Advanced cases, outside basic DSD
CAD/CAM	Computer-aided design / manufacturing	Milling or 3D printing of veneers, crowns, guides

8. Rules, edge cases and safety

8.1 Product rules

- **AI proposes, the dentist decides.** Every design must be editable, and the UI should label AI results as 'suggestion' until a clinician approves.
- **Never claim a diagnosis.** The software plans aesthetics. It does not diagnose gum disease, cavities, bone loss, or whether a treatment is advisable.
- **Show the measurement and the norm** side by side (e.g. 'Width/length 82% - target 75-80%'), with the norm configurable per clinic.
- **Expose uncertainty** (confidence scores). Allow manual drag-and-drop correction of every landmark. Re-compute dependent measurements automatically.
- **Before/after previews** must carry a visible note that results are simulations, not guaranteed outcomes.
- **Keep original photos immutable.** Edits are layers/versions.

8.2 Edge cases the AI must handle gracefully

Situation	What to do
Head tilted / rotated in the photo	Normalize using the pupil line. Warn if yaw/pitch is large (landmarks unreliable)
Smile does not show enough teeth	Request a better photo or mark the measurements as low confidence
Missing teeth, implants, existing crowns or bridges	Detect and mark as 'not natural'. Do not use them to compute 'natural' proportions
Braces / aligner attachments / heavy staining / dark gums	Reduce confidence, notify the user
Very crowded or rotated teeth	Flag: orthodontics may be better than veneers
Worn or short teeth (bruxism)	Flag for occlusion assessment. Do not just 'lengthen' them
Dentures	Different workflow: smile design for dentures uses other references (not covered here)
Children, mixed dentition (baby + adult teeth)	Out of scope for standard adult DSD. Warn
Ethnic and age diversity	Do not hard-code one 'ideal' smile. Offer presets (natural, youthful, bold) and let patient/clinician choose. Older faces show less incisor display naturally
Photos with phone wide-angle distortion	Estimate distortion or flag, as measurements near the image edge are off

8.3 Privacy and regulation

- Face photos and dental records are **health data**: use encryption in transit/at rest, access control and consent records.
- Obtain and store **explicit patient consent** for AI processing and for any use of images for training or marketing.
- Follow local rules (Egypt's personal data protection law, GDPR for EU patients, HIPAA for US). Medical-device regulation may apply if the tool claims to guide treatment decisions.
- Do not train models on patient data without a documented lawful basis and de-identification.

9. Glossary (A to Z)

Term	Meaning
Abutment	Tooth or implant post that supports a bridge or crown
Aligner	Clear removable plastic tray moving teeth
Anterior teeth	Front teeth: incisors and canines
Arch	The curved row of upper (maxillary) or lower (mandibular) teeth
Bonding	Gluing a restoration to the tooth with adhesive resin
Bruxism	Habitual grinding or clenching of teeth
Buccal	Toward the cheek
Canine (cuspid)	Pointed tooth at the corner of the smile
CBCT	3D X-ray used for bone/implant planning
Cervical	Neck of the tooth near the gum
Composite	Tooth-colored resin material for fillings and bonding
Contact point	Spot where two neighboring teeth touch
Crown	Visible part of the tooth, or a cap covering it
Diastema	Gap between teeth
Distal	Away from the midline along the arch
Embrasure	V-shaped space between two adjacent teeth
Enamel	Hard outer layer of the tooth
Facial/labial	Toward the face/lips
FDI	Two-digit international tooth numbering system
Gingiva	Gum
Gingival zenith	Highest point of the gum contour on a tooth
Gingivectomy	Surgical removal of gum tissue
Gummy smile	Excess gum visible when smiling
Incisal edge	Cutting edge of front teeth
Incisor	Front cutting teeth (central and lateral)
Interdental papilla	Gum triangle between two teeth
Intraoral scanner	Handheld 3D camera that digitally records teeth (replaces putty impressions)
Lingual / palatal	Toward the tongue / palate
Mandible / maxilla	Lower jaw / upper jaw
Mesial	Toward the midline along the arch
Mock-up	Temporary preview of the planned smile in the mouth
Occlusal plane	Imaginary plane where the upper and lower teeth meet
Occlusion	How upper and lower teeth meet/contact
Overbite / overjet	Vertical overlap / horizontal gap of upper vs lower incisors
Papilla	See interdental papilla
Prep (preparation)	Reducing a tooth to receive a veneer/crown
Prosthodontics	Dental specialty for replacing/restoring teeth (crowns, bridges, dentures, implants)
Restoration	Any filling, veneer, crown or other repair
Shade guide	Set of tooth-colored tabs for matching color
Smile arc	Curve of the upper incisal edges vs lower lip

Term	Meaning
Smile design (DSD)	Planning the smile on photos/video with digital tools
Veneer	Thin shell bonded to the front of a tooth
Wax-up	Sculpted preview model of the planned teeth (physical or digital)
Zirconia / e.max / feldspathic	Common ceramic materials for crowns/veneers (strength vs. translucency tradeoff)

Final checklist for the developer: (1) Normalize every photo to the pupil line. (2) Calibrate to millimeters and show uncertainty. (3) Store view type with every landmark. (4) Compute measurements, compare to configurable norms, explain the result. (5) Everything editable by the dentist. (6) Label simulations as simulations. (7) Protect patient data. When unsure about a clinical rule, ask the dentist: do not guess.